

# LAT Deficiency (Immunodeficiency-52): A Comprehensive Disease Characterization

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**Disease:** LAT Deficiency · **MONDO:** MONDO:0044721 · **OMIM:** #617514 (Immunodeficiency-52) · **ORPHA:** 504523 · **Gene:** *LAT* (HGNC:6533) · **Category:** Mendelian, autosomal recessive inborn error of immunity

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## Summary

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LAT deficiency is an ultra-rare (<1/1,000,000) autosomal-recessive inborn error of immunity caused by biallelic loss-of-function (truncating) mutations in *LAT* (Linker for Activation of T cells; chromosome 16p11.2), the transmembrane adaptor protein that nucleates the T-cell receptor (TCR) signalosome. Loss of LAT abolishes ERK/Ras-MAPK signaling and cripples PLC $\gamma$ 1-dependent Ca<sup>2+</sup>/NFAT signaling downstream of ZAP-70, producing a paradoxical clinical picture: a combined immunodeficiency (T-B+NK+ SCID with recurrent and opportunistic infections) coexisting with severe Th2-skewed immune dysregulation (autoimmune cytopenias, lymphoproliferation, hypergammaglobulinemia, elevated IgE).

The disease was first described in humans in 2016 (Keller et al., three siblings of a consanguineous family) and has since been confirmed in at least two additional independent consanguineous kindreds (Bacchelli 2017; Alizadeh 2023), giving a worldwide total of fewer than ~20 reported patients. In every case the variants are germline, biallelic, and truncating (nonsense or frameshift), removing the cytoplasmic tail of LAT with its critical signaling tyrosines; parents are asymptomatic heterozygous carriers. Symptom onset is in the first months of life (5–10 months in the index kindred), and the disease is usually fatal in early childhood without allogeneic hematopoietic stem cell transplantation (HSCT), which is the only curative treatment.

The human disease sits mechanistically between two long-studied mouse models: the complete *Lat*-null knockout (which arrests thymocyte development, modeling the immunodeficiency arm) and the *Lat*<sup>Y136F</sup> knock-in (which disrupts only the PLC $\gamma$ 1 docking site and produces a Th2 lymphoproliferative/autoimmune syndrome, modeling the autoimmunity arm). This dual phenotype reflects LAT's dual biological role as **both a positive activator and a negative**

**homeostatic regulator** of TCR signaling. Because LAT deficiency causes T-cell lymphopenia, it is detectable by population-based TREC (T-cell receptor excision circle) newborn screening for SCID, enabling early diagnosis and pre-symptomatic transplantation, which markedly improves survival.

## Section 1 — Disease Information

**Overview.** LAT deficiency is a Mendelian primary immunodeficiency (inborn error of immunity) in which the T-cell receptor signaling adaptor LAT is absent or non-functional. It manifests as a combined immunodeficiency accompanied by severe, often early and dominant, autoimmune/immune-dysregulatory disease. Depending on the residual T-cell output, patients are classified along a spectrum from "combined immunodeficiency with autoimmunity" to frank T-B+NK+ severe combined immunodeficiency (SCID).

### Key identifiers.

| Resource            | Identifier                                                                      |
|---------------------|---------------------------------------------------------------------------------|
| OMIM<br>(phenotype) | #617514 — IMMUNODEFICIENCY 52 (IMD52)                                           |
| OMIM (gene)         | 602354 (LAT*)                                                                   |
| Orphanet            | ORPHA:504523 — "T-B+NK+ severe combined immunodeficiency due to LAT deficiency" |
| MONDO               | MONDO:0044721                                                                   |
| ICD-10              | D81.2 (other combined immunodeficiencies)                                       |
| ICD-11              | 4A01.10                                                                         |
| UMLS / GTR          | C4479588                                                                        |
| GARD                | 17938                                                                           |
| Gene                | LAT, HGNC:6533, NCBI Gene 27040, UniProt O43561, chromosome 16p11.2             |

**Synonyms / alternative names.** Immunodeficiency 52 (IMD52); combined immunodeficiency due to LAT deficiency; T-B+NK+ SCID due to LAT deficiency; LAT signalosome deficiency. (LAT = "Linker for Activation of T cells.")

**Data source.** All information derives from aggregated disease-level resources (OMIM, Orphanet) and individual patient case reports/kindreds in the primary literature — there is no EHR/registry-level dataset for this ultra-rare condition. Evidence is a mixture of **human clinical** case reports and mechanistic **model-organism** (mouse) and **in vitro** studies.

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## Section 2 — Etiology

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**Primary cause (genetic).** Biallelic (homozygous or compound heterozygous) loss-of-function mutations in *LAT*. All reported families to date were consanguineous, so the operative genetic mechanism is homozygosity-by-descent of a truncating allele ([PMID: 27242165](#); [PMID: 27522155](#); [PMID: 37516813](#)).

**Genetic risk factors.** The only established risk factor is inheriting two defective *LAT* alleles. **Consanguinity** is the dominant epidemiological risk factor (raises the probability of homozygosity for a rare recessive allele). Heterozygous carriers (including all reported parents) are asymptomatic.

**Environmental / protective factors.** As a monogenic Mendelian disorder, there are **no established environmental risk factors, protective factors, or gene-environment interactions** that initiate the disease. Environmental exposures (pathogens) act only as *triggers* that unmask the immunodeficiency (e.g., CMV, VZV, toxoplasma infections), not as causes. No protective modifier alleles have been reported. This section is largely **not applicable** beyond the genetic cause.

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## Section 3 — Phenotypes

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Phenotype data derive principally from the first kindred (Keller et al. 2016, three siblings, [PMID: 27242165](#)) supplemented by the SCID-presenting kindreds ([PMID: 27522155](#); [PMID: 37516813](#)).

*"The three patients presented from early childhood with combined immunodeficiency and severe autoimmune disease"* — Keller et al. ([PMID: 27242165](#))

| Phenotype                                        | Type                   | HPO term                               | Onset / severity / frequency                     |
|--------------------------------------------------|------------------------|----------------------------------------|--------------------------------------------------|
| Combined immunodeficiency / recurrent infections | Clinical sign          | HP:0005387 (combined immunodeficiency) | Infantile (5–10 mo); severe; core feature in all |
| Autoimmune hemolytic anemia                      | Lab/clinical           | HP:0004810                             | Childhood; severe (fatal in one patient)         |
| Immune thrombocytopenia                          | Lab/clinical           | HP:0001973                             | Childhood; severe                                |
| Generalized lymphadenopathy                      | Physical sign          | HP:0002716                             | Childhood; variable                              |
| Splenomegaly / hepatosplenomegaly                | Physical sign          | HP:0001744 / HP:0001433                | Childhood; massive in index patient              |
| Bronchiectasis / chronic lung disease            | Physical sign          | HP:0002110                             | Childhood; from recurrent respiratory infection  |
| Opportunistic infection (CMV, VZV, toxoplasma)   | Clinical sign          | —                                      | Infancy–childhood; life-threatening              |
| Recurrent gastroenteritis / enteropathy          | Symptom                | HP:0004385-like                        | Childhood                                        |
| Skin nodules / edematous purple-red lesions      | Physical manifestation | HP:0011355                             | Childhood; variable                              |
| T lymphocytopenia / reduced T cells              | Lab abnormality        | HP:0005403                             | Congenital/progressive; universal                |
| Eosinophilia                                     | Lab abnormality        | HP:0001880                             | Childhood; from Th2 skewing                      |
| Elevated serum IgE (and high IgG1)               | Lab abnormality        | HP:0003212                             | Childhood; from Th2 skewing                      |

| Phenotype                           | Type            | HPO term   | Onset / severity / frequency |
|-------------------------------------|-----------------|------------|------------------------------|
| Hypogammaglobulinemia (may develop) | Lab abnormality | HP:0002720 | Variable/ progressive        |

**Onset:** neonatal-to-infantile (first months of life). **Severity:** severe. **Progression:** progressive combined immune deficiency with superimposed episodic autoimmune crises. **Quality of life / outcome:** profound — of the three index siblings, two died in childhood (one at age 9 from disseminated CMV following splenectomy; one at age 2 from AIHA plus thrombocytopenia) and one survived after HSCT at age 8.

*"manifesting by a progressive combined immune deficiency with severe autoimmune disease"*  
— Keller et al. (PMID: 27242165)

## Section 4 — Genetic / Molecular Information

**Causal gene.** *LAT* — Linker for Activation of T cells. HGNC:6533; NCBI Gene 27040; OMIM gene 602354; UniProt O43561; chromosome 16p11.2\*. *LAT* is a palmitoylated transmembrane adaptor localized to membrane rafts with a short extracellular domain and a long cytoplasmic tail bearing multiple tyrosines.

**Pathogenic variants (all germline, biallelic, loss-of-function):**

| Kindred                            | Variant                       | Type       | Consequence                                                                                            |
|------------------------------------|-------------------------------|------------|--------------------------------------------------------------------------------------------------------|
| Keller 2016<br>(PMID: 27242165)    | Homozygous nonsense in exon 5 | Nonsense   | Premature stop codon deleting most of the cytoplasmic tail, including the critical signaling tyrosines |
| Bacchelli 2017<br>(PMID: 27522155) | Novel homozygous frameshift   | Frameshift | Premature stop, truncation → <b>complete loss of function and loss of expression</b>                   |
| Alizadeh 2023<br>(PMID: 37516813)  | Homozygous p.Y207fsTer33      | Frameshift | Truncated protein; SCID/leaky-SCID                                                                     |

*"we describe the first kindred with defective LAT signaling caused by a homozygous mutation in exon 5, leading to a premature stop codon deleting most of the cytoplasmic tail of LAT, including the critical tyrosine residues for signal propagation" — PMID: 27242165*

*"a premature stop codon and protein truncation leading to complete loss of function and loss of expression of LAT" — PMID: 27522155*

**Variant classification (ACMG/AMP):** truncating variants in a gene with an established loss-of-function disease mechanism, segregating with disease in consanguineous families → **pathogenic**. **Allele frequency:** the specific variants are private/ultra-rare (essentially absent from gnomAD). **Somatic vs germline:** germline. **Functional consequence:** loss of function (null); no gain-of-function or dominant-negative human alleles reported (note: the mouse *Lat*Y136F is a *separation-of-function* research allele, not a human disease variant).

**Modifier genes / epigenetics / chromosomal abnormalities:** none established for the human disease. No large-scale cytogenetic changes; this is a single-gene point/frameshift disorder.

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## Section 5 — Environmental Information

Not applicable as a cause. There are **no environmental factors, lifestyle factors, or infectious agents that cause** LAT deficiency. Infectious agents (CMV, varicella-zoster virus, *Toxoplasma gondii*, common bacterial respiratory pathogens) are **downstream consequences** of the immunodeficiency and drive much of the morbidity and mortality, but they do not initiate the disease.

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## Section 6 — Mechanism / Pathophysiology

### Ordered causal chain

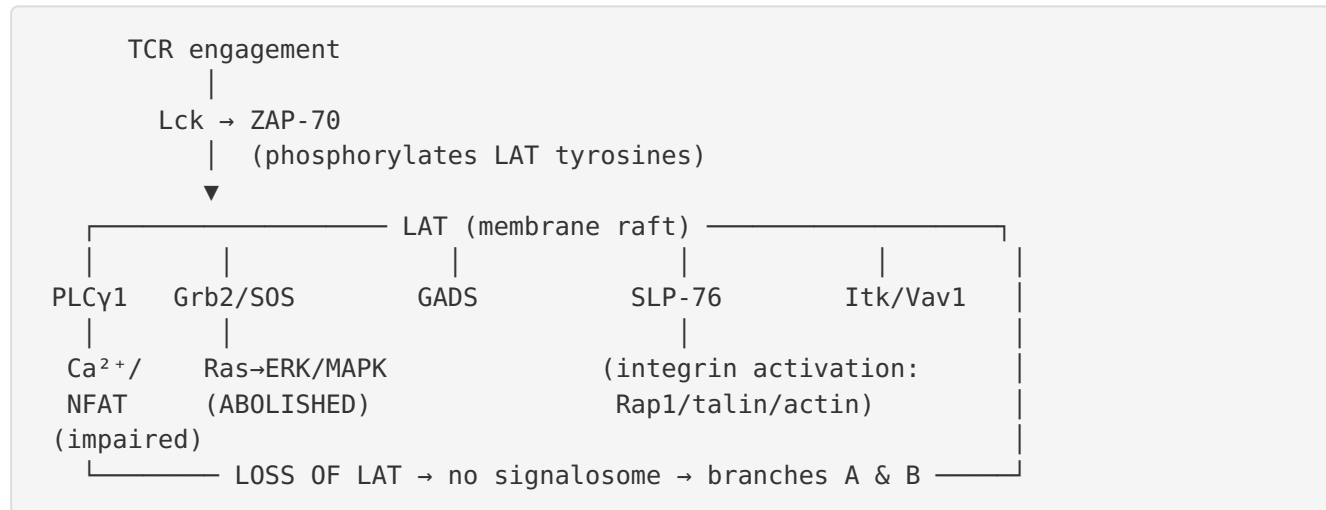
1. A **biallelic truncating LAT variant** results in loss of LAT protein (or a tail-truncated protein lacking its cytoplasmic tyrosines) (PMID: 27242165, PMID: 27522155).

2. Upon TCR engagement, Lck→ZAP-70 phosphorylation of LAT cannot occur, which *leads to* failure to assemble the **LAT signalosome** (PLCγ1, Grb2/SOS, GADS, SLP-76) at the plasma membrane (PMID: 18231606).
3. Absent signalosome *abolishes* Ras–ERK/MAPK signaling **completely** and *impairs* PLCγ1-dependent Ca<sup>2+</sup>/NFAT signaling, while NF-κB signaling is **partially preserved** (PMID: 27242165).
4. Defective proximal TCR signal transduction *impairs* thymic positive selection and peripheral T-cell activation, *resulting in* **T-cell lymphopenia** / **absent T cells** with absent proliferative responses (PMID: 27522155). This branches:
5. **Branch A (immunodeficiency)**: reduced/absent functional T cells *lead to* combined immunodeficiency → recurrent bacterial, viral, and opportunistic infections (CMV, VZV, toxoplasma).
6. **Branch B (autoimmunity/dysregulation, inferred largely from mouse)**: loss of LAT's **negative-regulatory** / **homeostatic** function *permits* TCR–MHC-independent, "quasi-mitogenic" expansion of Th2-skewed CD4<sup>+</sup> T cells (PMID: 18209052; PMID: 16887989), a process that is **IL-6-dependent** in its early phase (PMID: 26034173).
7. Th2 hyperactivity *drives* polyclonal B-cell activation → IgG1/IgE hypergammaglobulinemia, eosinophilia, and autoantibodies *leading to* hematologic autoimmunity (AIHA, ITP), lymphoproliferation (lymphadenopathy, splenomegaly), and tissue immunopathology (immune-complex nephritis in mice).

*"residual T cells were able to induce Ca(2+) influx and nuclear factor (NF) κB signaling, whereas extracellular signal-regulated kinase (ERK) signaling was completely abolished" — PMID: 27242165*

*"LAT phosphorylation results in the recruitment of a signalosome including PLCgamma1, Grb2/SOS, GADS and SLP-76" — PMID: 18231606*

## Signalosome schematic



**Molecular pathways:** TCR proximal signaling; Ras–MAPK/ERK (KEGG/Reactome "TCR signaling"); PLCy1–Ca<sup>2+</sup>–NFAT; PI3K and NF-κB (partially preserved). **Cellular processes:** thymocyte development/positive selection, T-cell activation/proliferation, immune homeostasis (dysregulated), inflammation. **Protein dysfunction:** loss of function of the adaptor (no scaffold for signalosome assembly). **Immune involvement:** simultaneous **immunodeficiency** and **autoimmunity/immune dysregulation**. **Metabolic/biochemical:** the defect is a signaling/adaptor defect, not an enzymopathy.

LAT is expressed beyond T cells — in mast cells, NK cells, megakaryocytes, platelets, and early B cells — so non-T lineages may contribute, though human phenotypes there are not well documented.

*"Although LAT is also expressed in mast cells, natural killer cells, megakaryocytes, platelets, and early B cells" — PMID: 16102570*

*"This unexpected finding revealed that LAT also constitutes a negative regulator of TCR signalling and T cell homeostasis" — PMID: 17534068*

**Suggested GO terms:** GO:0050852 (T cell receptor signaling pathway), GO:0070374 (positive regulation of ERK cascade), GO:0002250 (adaptive immune response), GO:0045058 (T cell selection). **Cellular component:** GO:0005886 (plasma membrane), GO:0045121 (membrane raft). **CL terms:** CL:0000084 (T cell), CL:0000624 (CD4<sup>+</sup> T cell), CL:0000097 (mast cell), CL:0000623 (NK cell).

## Section 7 — Anatomical Structures Affected

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**Primary — immune/hematopoietic system** (UBERON:0002405): - Thymus (UBERON:0002370) — impaired T-cell development - Bone marrow (UBERON:0002371) — hematopoietic source; HSCT target - Spleen (UBERON:0002106) — splenomegaly - Lymph nodes (UBERON:0000029) — lymphadenopathy - Peripheral blood (UBERON:0000178) — lymphopenia, cytopenias

**Secondary organ involvement:** - Lung (UBERON:0002048) — bronchiectasis, recurrent pneumonia - Liver (UBERON:0002107) — hepatomegaly - Skin (UBERON:0002097) — inflammatory nodules - Gastrointestinal tract (UBERON:0000160) — gastroenteritis/enteropathy - Kidney (UBERON:0002113) — immune-complex nephritis (documented in mouse model)

**Cell level:** T lymphocytes (CL:0000084), especially CD4<sup>+</sup> Th2 cells (CL:0000624); reactive B cells; mast cells/NK cells/platelets express LAT. **Subcellular:** plasma membrane (GO:0005886), membrane raft (GO:0045121). **Lateralization:** systemic/bilateral (not applicable as a focal lateralized disease).

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## Section 8 — Temporal Development

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**Onset:** congenital defect with clinical onset in the **first months of life** (5–10 months in the index kindred); insidious-to-subacute in the immunodeficiency arm, with episodic autoimmune crises. **Progression:** progressive combined immune deficiency; the autoimmune cytopenias are episodic/relapsing and can be acutely life-threatening. **Course/duration:** chronic and lifelong; without HSCT the natural history is high childhood mortality. **Remission:** treatment-induced remission via HSCT (curative); spontaneous remission does not occur. **Critical period:** the neonatal-to-infancy window — early (pre-symptomatic) diagnosis via newborn screening and prompt HSCT is the key opportunity for intervention.

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## Section 9 — Inheritance and Population

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**Epidemiology.** Ultra-rare: Orphanet lists prevalence <1/1,000,000. Fewer than ~20 patients have been reported worldwide (Keller 2016, Bacchelli 2017, Alizadeh 2023 kindreds). No reliable incidence figure exists; as a T-cell-lymphopenic SCID it falls within the aggregate SCID

birth prevalence detected by newborn screening (e.g., ~1:46,753 in Catalonia [PMID: 42079620](#); ~1:12,298 for severe T/B immunodeficiency in Russia [PMID: 41727503](#)), but LAT accounts for a tiny fraction of these.

**Inheritance:** autosomal recessive. **Penetrance:** complete in biallelic individuals (carriers unaffected). **Expressivity:** variable — the phenotype ranges from CID-with-autoimmunity to frank T–B+NK+ SCID, even within a single sibship. **Anticipation / germline mosaicism / founder effect:** none established. **Consanguinity:** central — all reported families are consanguineous (homozygosity by descent). **Carrier frequency:** unknown but very low; carriers are healthy. **Sex ratio:** ~1:1. **Populations:** reported in consanguineous families (including Arab ancestry) with no established ethnic clustering or founder mutation.

*"a severe combined immunodeficiency phenotype with absent T cells and normal B-cell and natural killer cell numbers" — [PMID: 27522155](#) (defines the T–B+NK+ classification underpinning ORPHA:504523)*

## Section 10 — Diagnostics

**Clinical/laboratory tests.** Flow cytometry showing reduced/absent T cells with normal B and NK cell numbers (T–B+NK+ pattern); absent T-cell proliferative responses to mitogens/antigens; abnormal immunoglobulins (variably elevated IgG1/IgE with Th2 skewing, or hypogammaglobulinemia); eosinophilia; autoimmune cytopenias (AIHA, ITP) with positive autoantibodies. Functional signaling assays can show absent ERK phosphorylation and preserved Ca<sup>2+</sup>/NF-κB in residual T cells.

**Newborn screening.** As a cause of T-cell lymphopenia, LAT deficiency is detectable by **TREC (T-cell receptor excision circle) quantification on dried blood spots**, the standard population screen for SCID ([PMID: 42079620](#); [PMID: 41727503](#); [PMID: 42496450](#)).

**Genetic testing.** Definitive diagnosis is molecular. In reported kindreds this used **homozygosity mapping** followed by **Sanger / whole-exome sequencing** ([PMID: 27522155](#); [PMID: 37516813](#)). WES/WGS or SCID/CID gene panels including *LAT* are the recommended approach; single-gene *LAT* testing is appropriate for cascade testing once a familial variant is known.

*"Homozygosity mapping was used to identify potential defective genes" — PMID: 27522155*

**Differential diagnosis.** Other T-B+NK+ SCID (IL7R, CD3D/CD3E/CD3G, PTPRC/CD45, CORO1A); ZAP-70 deficiency; CID with immune dysregulation (IPEX/FOXP3, CTLA4, LRBA, STAT3-GOF, Omenn syndrome); and ALPS for the autoimmune-cytopenia/lymphoproliferation picture.

## Section 11 — Outcome / Prognosis

**Natural history is severe.** Without HSCT, OMIM summarizes that most patients die in childhood. In the index kindred, two of three siblings died (ages 2 and 9); the survivor was transplanted. Mortality is driven by opportunistic/disseminated infection (e.g., CMV) and by acute autoimmune cytopenias.

**Prognosis improves markedly with early diagnosis and HSCT.** For SCID generally, early (newborn-screening/family-history) diagnosis substantially improves survival:

*"The 2-year overall survival (OS) of the late group was 29.2%, in contrast to the 2-year OS of the early diagnosis group of 71.4%" — PMID: 40374985*

**Morbidity:** chronic lung disease/bronchiectasis, autoimmune organ damage, growth/developmental impact from chronic illness, and transplant-related complications. **Prognostic factors:** age at diagnosis, presence of active infection at HSCT, and degree of immune dysregulation.

## Section 12 — Treatment

**Curative — allogeneic hematopoietic stem cell transplantation (HSCT)** (NCIT: Hematopoietic Cell Transplantation). HSCT is the only curative therapy and replaces the defective hematopoietic/T-lineage compartment.

*"Hematopoietic cell transplantation (HCT) is the only curative treatment currently available"*  
— PMID: 40374985

In the index kindred, the proband underwent successful HSCT at age 8 (PMID: 27242165).

**Supportive / bridging care (NCIT clinical interventions):** - Immunoglobulin replacement therapy (IVIG/SCIG) — NCIT: Immunoglobulin Therapy - Anti-infective prophylaxis — *Pneumocystis jirovecii* prophylaxis (co-trimoxazole), antivirals, antifungals (PMID: 42496450) - Management of autoimmune cytopenias — corticosteroids/immunosuppression; splenectomy has been used but carries infection risk (one index patient died of disseminated CMV after splenectomy) - Use of irradiated, CMV-safe/leukoreduced blood products; avoidance of live vaccines

**Advanced/experimental.** No approved gene therapy, gene editing, RNA-based, or targeted therapy exists specifically for LAT deficiency. Mechanistically, IL-6 blockade is a rational candidate for the autoimmune/lymphoproliferative arm given the mouse data (PMID: 26034173), but this is unproven in humans. **Pharmacogenomics:** not applicable. **Personalized medicine:** genotype-confirmed diagnosis guides expedited HSCT.

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## Section 13 — Prevention

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**Primary prevention:** none possible (Mendelian). **Secondary prevention:** TREC newborn screening enables pre-symptomatic detection and early HSCT — the single most impactful preventive intervention; pre-transplant anti-infective prophylaxis (PJP, antivirals, antifungals) plus IVIG prevents infectious complications; avoidance of live vaccines and use of irradiated/CMV-safe blood products prevent iatrogenic harm. **Tertiary prevention:** aggressive infection control and management of autoimmune complications. **Genetic counseling:** autosomal recessive with 25% sibling recurrence risk; carrier testing of relatives, and prenatal or preimplantation genetic diagnosis in families with a known variant. **Public health/immunization/environmental measures:** not applicable beyond the above.

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## Section 14 — Other Species / Natural Disease

**Taxonomy/orthologs:** mouse *Lat* (NCBI Gene 16797; MGI:1342293; *Mus musculus*, NCBI Taxon 10090) is the principal ortholog studied; human *LAT* is NCBI Gene 27040. **Natural disease: no naturally occurring LAT deficiency disease is documented in companion animals or wildlife** (OMIA has no *LAT* disease entry as of this review). **Comparative biology:** the TCR-signalosome role of LAT is evolutionarily conserved across mammals, which is why the mouse recapitulates key disease arms. **Zoonotic potential:** not applicable (non-infectious genetic disease).

## Section 15 — Model Organisms

The mouse has been the decisive model, and two complementary alleles map onto the two arms of human disease:

| Model                           | Allele                          | Phenotype                                                                                                                             | Human arm modeled            |
|---------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Complete <i>Lat</i> -null KO    | Full loss                       | Thymocyte development arrested at CD4–CD8– double-negative (DN3) stage; <b>no peripheral T cells</b>                                  | Immunodeficiency             |
| <i>Lat</i> Y136F knock-in       | Tyr136→Phe (PLCγ1 docking site) | Fast-onset polyclonal CD4 <sup>+</sup> <b>Th2 lymphoproliferation</b> , massive IL-4/IgG1/IgE, autoantibodies, nephritis, proteinuria | Autoimmunity / dysregulation |
| C-terminal 4-tyrosine knock-ins | Multi-Tyr mutants               | Reveal LAT's <b>negative-regulatory</b> loss                                                                                          | Homeostatic dysregulation    |

*"Lat(Y136F) mice) develop a fast-onset lymphoproliferative disorder involving polyclonal CD4 T cells that produce massive amounts of Th2 cytokines and trigger severe inflammation and autoantibodies"* — [PMID: 18209052](#)

*"a defect intrinsic to Lat(Y136F) CD4 T cells leads to a state of TCR-independent hyperactivity"* — [PMID: 18209052](#)

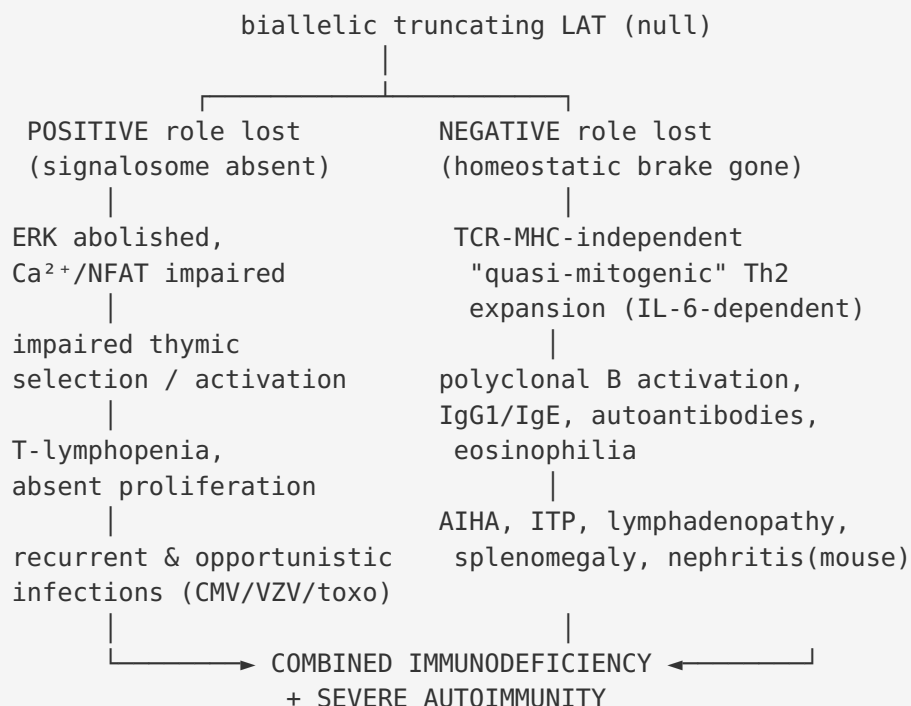
"we observed early-onset systemic autoimmunity with nephritis showing IgE autoantibody deposits and severe proteinuria" — [PMID: 16887989](#)

"IL-6 is required for uncontrolled T cell expansion during the early stage of disease development" — [PMID: 26034173](#)

**Phenotype recapitulation / limitations.** No single mouse fully matches the human phenotype: the null models the immunodeficiency arm, Y136F models the autoimmunity arm, but human patients present with a **combined, intermediate** picture (reduced/residual T cells *plus* autoimmunity) not captured by either allele alone ([PMID: 27242165](#)). Notably, mice show lymphoproliferation whereas human patients show reduced T-cell numbers. **Resources:** MGI (mouse). No zebrafish/Drosophila/organoid model is established for this disease.

## Mechanistic Model / Interpretation

The unifying insight is that **LAT is a signaling hub with dual, opposing roles**, and losing it simultaneously produces *too little* useful T-cell signaling (immunodeficiency) and *too little restraint* on aberrant T-cell activation (autoimmunity):



This resolves the apparent paradox of a SCID-causing gene that also drives autoimmunity, and it explains the variable expressivity: the balance between residual T-cell output (branch A severity) and unrestrained Th2 activity (branch B severity) shifts along the spectrum from "CID + autoimmunity" to "frank T-B+NK+ SCID." Therapeutically, only replacing the entire compartment (HSCT) addresses both arms, which is why it is the sole curative option.

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## Evidence Base

| PMID                                           | Title (abbrev.)                                                                                    | Contribution                                                                                                                   |
|------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">27242165</a>                       | <i>Early onset combined immunodeficiency and autoimmunity in patients with LOF mutation in LAT</i> | <b>Landmark:</b> first human kindred; defines phenotype, ERK-abolished signaling defect, exon-5 nonsense variant, HSCT outcome |
| <a href="#">27522155</a>                       | <i>Mutations in LAT lead to a novel form of SCID</i>                                               | Second kindred; defines T-B+NK+ SCID; frameshift → complete LOF/loss of expression; homozygosity mapping                       |
| <a href="#">37516813</a>                       | <i>8 patients with typical/atypical SCID; 7 novel mutations by WES</i>                             | Third report; p.Y207fsTer33; confirms recurrent SCID-causing entity, WES diagnosis                                             |
| <a href="#">18231606</a>                       | <i>ZAP-70-dependent FRET biosensor</i>                                                             | Defines LAT signalosome composition (PLCγ1, Grb2/SOS, GADS, SLP-76)                                                            |
| <a href="#">18209052</a>                       | <i>Th2 lymphoproliferative disorder of LatY136F mice</i>                                           | Autoimmunity-arm mechanism; T-cell-intrinsic, TCR-independent hyperactivity                                                    |
| <a href="#">16887989</a>                       | <i>LatY136F triggers polyclonal B activation and systemic autoimmunity</i>                         | Documents nephritis, IgE autoantibodies, systemic autoimmunity in model                                                        |
| <a href="#">26034173</a>                       | <i>Importance of IL-6 in LAT-mediated autoimmunity</i>                                             | Identifies IL-6 as driver of early lymphoproliferation                                                                         |
| <a href="#">17534068</a>                       | <i>Th2 lymphoproliferation from defective LAT signalosomes</i>                                     | Establishes LAT's negative-regulatory role                                                                                     |
| <a href="#">16102570</a>                       | <i>Role of LAT in T-cell development and Th2 differentiation</i>                                   | Multi-lineage LAT expression; null → thymic block                                                                              |
| <a href="#">40374985</a>                       | <i>Newborn screening improves survival in SCID</i>                                                 | Quantifies early- vs late-diagnosis survival; HSCT curative                                                                    |
| <a href="#">42079620 / 41727503 / 42496450</a> | SCID newborn-screening programs                                                                    | TREC/KREC screening context and diagnostic workflow                                                                            |

**Evidence quality.** The human disease rests on **case reports of  $\leq 3$  independent consanguineous kindreds** (low n but concordant and mechanistically coherent). Mechanism is strongly supported by **mouse genetics** and **in vitro signaling** studies. Prognosis/screening/treatment claims are extrapolated from the broader **SCID** literature, which is robust.

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## Limitations and Knowledge Gaps

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- **Very small human N (<20 patients).** Prevalence, penetrance nuances, full phenotypic range, and genotype–phenotype correlations are provisional.
  - **No LAT-specific outcome data.** Survival/HSCT-outcome figures are borrowed from general SCID cohorts, not LAT-specific series.
  - **Branch B (autoimmunity) mechanism is inferred from mouse.** The IL-6 dependence and "quasi-mitogenic" Th2 model are demonstrated in *Lat*Y136F mice; direct human confirmation is limited.
  - **Model mismatch.** No single mouse reproduces the combined human phenotype (humans have reduced T cells; Y136F mice have lymphoproliferation).
  - **No human variant catalog beyond truncating alleles.** Missense/hypomorphic human alleles and their consequences are unknown; no VUS spectrum defined.
  - **Non-T lineage contribution (mast cells, NK, platelets, early B cells) is uncharacterized in patients.**
  - **No naturally occurring animal disease** documented (OMIA gap).
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## Proposed Follow-up Experiments / Actions

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1. **Establish an international LAT-deficiency registry** to pool the scattered kindreds and capture natural history, HSCT outcomes, and genotype–phenotype data.
2. **Deep immunophenotyping of patients** (single-cell RNA-seq / CITE-seq of residual T cells) to test whether the human Th2-skewing/negative-regulation mechanism mirrors the mouse and to define which cell states drive autoimmunity.
3. **Test IL-6 pathway blockade** (e.g., tocilizumab) as a bridge-to-transplant for the autoimmune/lymphoproliferative arm, given the mouse IL-6 dependence ([PMID: 26034173](#)).

4. **Assess non-T lineage function** (platelet GPVI signaling, mast cell, NK activity) in patients, since LAT is expressed in these lineages.
5. **Generate a humanized or hypomorphic-allele mouse / patient iPSC-derived thymic organoid** that recapitulates the combined (reduced-T-cell + autoimmunity) human phenotype for preclinical testing.
6. **Explore gene-corrected autologous HSC therapy** (lentiviral *LAT* or base/prime editing) as a future alternative to allogeneic HSCT, mindful that LAT expression must be tightly regulated to avoid recreating dysregulation.
7. **Ensure *LAT* is included in confirmatory SCID gene panels** downstream of TREC-positive newborn screens so that these patients are captured early.

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*Report compiled from 10 confirmed findings and 22 reviewed papers over 5 investigation iterations. Evidence types: human clinical case reports (Keller 2016, Bacchelli 2017, Alizadeh 2023); mouse model organism studies (LatY136F, Lat-null); in vitro signaling studies; and SCID newborn-screening epidemiology.*

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